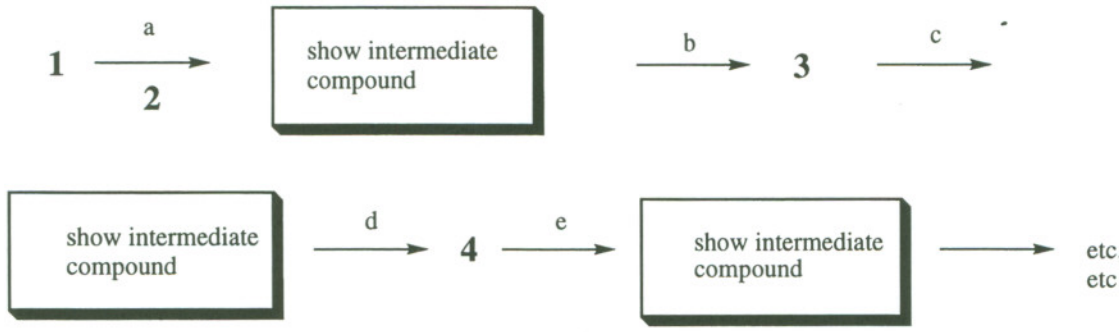
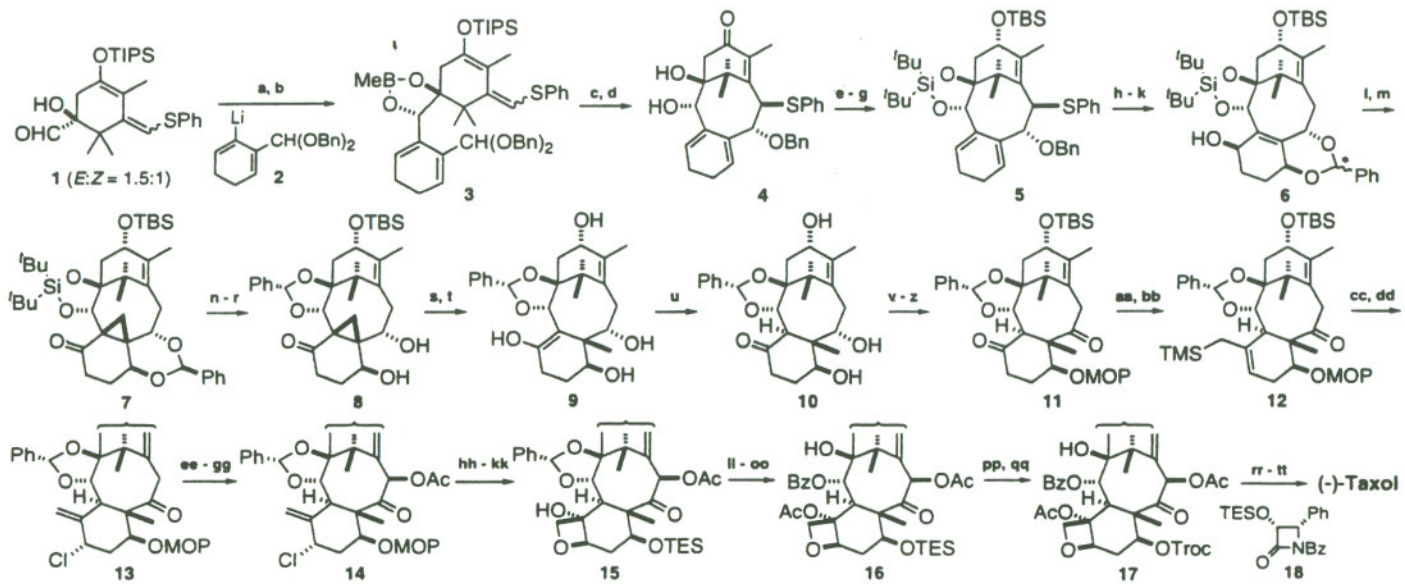
Organic Division Cumulative Exam January 16, 1999

Draw the products of each of the reactions shown in the scheme, i.e. draw the product resulting from reaction of structure **1** with reagents **a**, then with reagents **b** to give structure **3**, etc, as indicated below. Be sure to draw the complete structure for each intermediate compound with the correct stereochemistry. You do <u>not</u> need to give the mechanisms of the reactions.



Scheme 1[a]



[a] ^{t}BuMgCl, THF, −78 °C; then **2**, −78 °C, 68%. (b) $(MeBO)_3$, pyr, benzene, 77%. (c) $TiCl_2(O^iPr)_2$, CH_2Cl_2, −78−0 °C. (d) Pinacol, DMAP, benzene, 59% from **3**. (e) BuLi, ^{t}Bu$_2$Si(H)Cl, THF, −78−0 °C. (f) DIBAL, CH_2Cl_2, −78 °C. (g) TBSOTf, 2,6-lutidine, CH_2Cl_2, −23 °C, 62% from **4**; (h) O_2, TPP, hv, CH_2Cl_2. (i) Bu_3SnH, AIBN, benzene, reflux, 80% from **5**. (j) Pd/C, H_2, EtOH. (k) $PhCH(OMe)_2$, CSA, CH_2Cl_2, −23 °C, 79% in 2 steps (α:β = ca. 1:4). (l) Et_2Zn, $ClCH_2I$, toluene, 0 °C, 66%. (m) Dess−Martin, CH_2Cl_2, 77%. (n) $Pd(OH)_2$, H_2, EtOH, 84%. (o) Triphosgene, pyr, CH_2Cl_2, −45 °C. (p) TBAF, AcOH, THF, 95% in 2 steps. (q) $PhCH(OMe)_2$, PPTS, benzene, reflux, 86%. (r) K_2CO_3, MeOH-THF, quant. (s) SmI_2, THF−MeOH−HMPA, quant. (t) TBAF, BHT, THF. (u) NaOMe, BHT, degassed MeOH, 45% in 2 steps (ca 50% of **9** was recovered); 65% by repeating this isomerization procedure twice. (v) $PhB(OH)_2$, CH_2Cl_2; (w) TBSOTf, 2,6-lutidine, CH_2Cl_2, −45 °C; (x) H_2O_2, $NaHCO_3$, AcOEt, 70% from **10**. (y) Dess−Martin, CH_2Cl_2, 92%. (z) 2-Methoxypropene, PPTS, CH_2Cl_2, 97%. (aa) KHMDS, $PhNTf_2$, −78 °C, 89%; (bb) $Pd(PPh_3)_4$, $TMSCH_2MgCl$, Et_2O, 91%; (cc) NCS, MeOH, 88%; (dd) 2-methoxypropene, PPTS, CH_2Cl_2, 89%. (ee) LDA, MoOPH, THF, −23 °C, 80%. (ff) Ac_2O, DMAP, CH_2Cl_2, 92%. (gg) DBN, toluene, reflux, 68% at 92% conversion. (hh) OsO_4, pyr, Et_2O, 86%. (ii) DBU, toluene, reflux, 86%. (jj) PPTS, MeOH. (kk) TESCl, imidazole, DMF, 97% in 2 steps. (ll) $Pd(OH)_2$, H_2, EtOH, 97%. (mm) triphosgene, pyr, CH_2Cl_2, −78−0 °C, 94%. (nn) Ac_2O, DMAP, CH_2Cl_2, 66%. (oo) PhLi, THF, −78 °C, 83%. (pp) HF·pyr, THF, 88%. (qq) trocCl, pyr, CH_2Cl_2, 94%. (rr) TASF, THF, 80%. (ss) LHMDS, **18**, THF, −78−0 °C, 77% at 90% conversion. (tt) Zn, AcOH-H_2O, 84%.